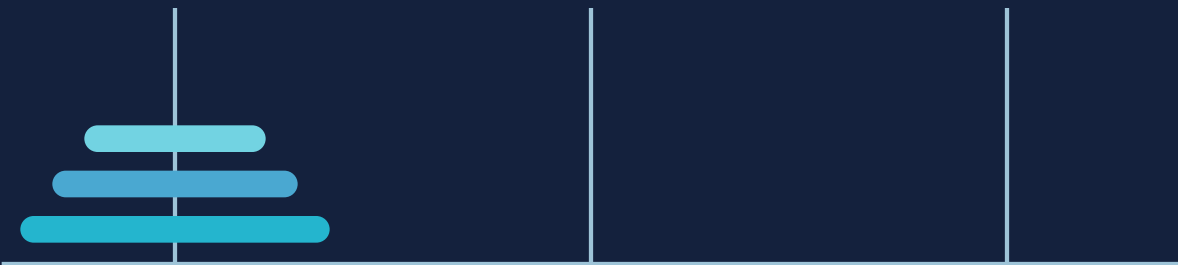


Robotics Research

Research Project

GPU-Accelerated Visual Verification and Recovery for
Long-Horizon Robot Manipulation



A controlled test of reliability under uncertainty

RESEARCH PROJECT

Contents

A closed-loop manipulation study of visual state reconciliation, bounded recovery, and reproducible sim-to-real reliability.

01	Executive research contract Thesis, primary hypothesis, contributions, and scope	1
02	Closed-loop system architecture Perception, state reconciliation, verification, and recovery	2
03	Technical method Symmetry-aware perception, planning, and residual alignment	3
04	Reference implementation Hardware baseline, software stack, and compatibility gates	4
05	Experimental design Comparisons, disturbances, protocol, and success criteria	5
06	Sixteen-week execution plan Milestones, go/no-go gates, and minimum viable completion	6
07	Risks, observability, and reproducibility Failure taxonomy, run-level evidence, and deliverables	7
08	Final demonstration and portfolio framing Demonstration sequence, evidence, and results communication	8
09	References and decision record Technical sources, compatibility notes, and decision log	9

1. Executive research contract

Thesis, hypothesis, contributions, and scope boundaries for a defensible robotics study.

Project thesis

Hanoi-RX studies whether closed-loop visual verification and a learned residual alignment policy can improve the reliability of long-horizon robot manipulation under calibration error, object displacement, and grasp or placement uncertainty.

Tower of Hanoi is the evaluation environment. The contribution is the system's ability to reconcile expected and observed state, identify when execution diverges, and select a bounded correction without surrendering safety to an end-to-end learned policy.

Primary hypothesis

Under preregistered physical disturbances, visual verification plus residual recovery will improve seven-move puzzle completion by at least 25 percentage points over the same manipulation stack operated open-loop, while producing zero illegal symbolic moves and zero collisions in the final battery.

Research contributions

- 1A closed-loop state-reconciliation architecture that maintains expected state, observed state, and confidence separately.
- 2A symmetry-aware perception benchmark comparing simple RGB-D geometry against an accelerated pose-estimation alternative.
- 3A vision-conditioned residual alignment policy trained in GPU-parallel simulation for the final centimeters of grasp or peg placement.
- 4A reproducible sim-to-real evaluation with controlled disturbances, ablations, latency profiles, and failure accounting.

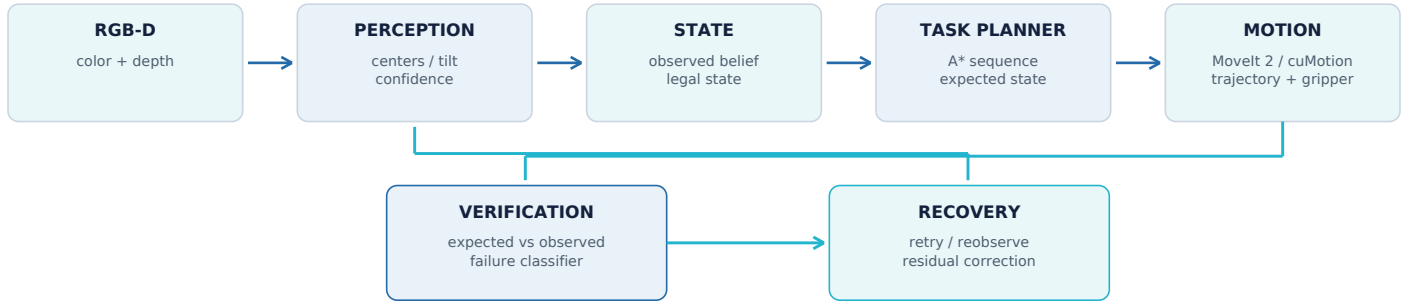
Scope boundary

In scope	Explicitly out of scope
Three-disk closed-loop completion; arbitrary legal starts; one learned residual policy; one RGB-D camera; single arm; controlled disturbances; Jetson inference benchmark.	Natural-language control; general-purpose VLA training; multi-camera fusion; mobile manipulation; five-disk completion; replacing the task planner with reinforcement learning.

Success is a measured result, not a flawless demo. A negative result remains valuable if the benchmark, failure analysis, and reproducibility are strong.

2. Closed-loop system architecture

The robot never assumes that a commanded action succeeded; every placement closes the observation loop.



Every placement returns through observation, reconciliation, verification, and bounded recovery.

State contract

The system stores three distinct objects: expected state from the planner, observed state from perception, and a belief state that reconciles evidence without hiding ambiguity.

```
ObservedState(disks={id: center_xyz, radius, peg, height, tilt, confidence},
board_frame=T_world_board, timestamp=t)
```

```
Decision = CONTINUE | REOBSERVE | RETRY | RESIDUAL_CORRECT | REPLAN | SAFE_STOP
```

Safety and authority boundaries

- The symbolic planner alone determines legal disk moves.
- MoveIt 2 or cuMotion owns gross collision-free motion; the learned policy cannot generate unconstrained workspace trajectories.
- The learned policy is bounded to a small Cartesian correction envelope near the grasp or placement target.
- Low confidence, repeated disagreement, controller faults, or workspace violations cause a safe stop rather than an improvised action.
- All state transitions and recovery decisions are logged with inputs, confidence, latency, and outcome.

Planner choice

Use the recursive Hanoi solver only as a canonical-state correctness oracle. Use a single A* state-space planner for arbitrary legal starts and replanning. PDDL and additional planners are unnecessary unless the project later expands beyond Hanoi.

3. Technical method

A symmetry-aware perception baseline, classical task and motion planning, and a bounded learned residual.

3.1 Symmetry-aware perception

A Hanoi disk is rotationally symmetric about its central axis. Full orientation around that axis is unobservable and irrelevant. The operational state therefore uses disk center, radius or identity, peg assignment, stack height, tilt, and confidence - not an artificial full 6-DoF target.

Method	Role	Evaluation
RGB-D geometry	Primary baseline: crop board, segment depth/color, fit circles or annuli, estimate centers and stack heights.	Accuracy, latency, sensitivity to lighting and occlusion.
FoundationPose	Accelerated comparison when CAD-based tracking adds value; use symmetry handling and lower-frequency estimation with tracking.	Pose/state accuracy, failure modes, GPU memory, p50/p95 latency.
AprilTag board frame	Stable board registration and calibration diagnostic; not used to identify individual disks.	Frame error and drift over time.

3.2 Classical planning and motion

- A* searches legal symbolic states and emits source-peg, destination-peg actions.
- A grasp generator converts each symbolic action into pre-grasp, approach, grasp, lift, transfer, place, release, and retreat phases.
- MoveIt 2 is the required baseline. cuMotion is a measured alternative, not a critical-path dependency.
- Collision geometry includes the arm, gripper, table, board, pegs, disk stacks, and a configurable human exclusion volume.
- Post-action observation begins only after controller completion and a short mechanically justified settling interval.

3.3 Learned residual alignment

The learned component corrects only the final alignment near a grasp or peg placement. It receives a local depth/RGB crop, nominal target transform, recent controller error, and confidence. It returns a clipped Cartesian residual and an abstain score.

Element	Design decision
Training	GPU-parallel Isaac Lab environments with randomized board pose, camera extrinsics, depth noise, friction, disk offset, and gripper alignment.
Action	Bounded delta-x, delta-y, delta-z and optional small tilt correction; no global trajectory generation.
Objective	Successful centered grasp or collision-free peg insertion with minimum correction magnitude and time.
Fallback	If the policy abstains or its confidence is below threshold, use rule-based reobservation/retry or safe stop.
Ablation	Fixed nominal target versus hand-coded correction versus learned residual correction.

4. Reference implementation

A compatibility baseline to validate before substantial hardware spending or research development.

The configuration below is a compatibility baseline, not a purchase mandate. The first two weeks must validate the exact arm, gripper, camera, drivers, and pinned software branch before substantial hardware spending.

Reference hardware

Component	Reference choice	Reason / decision gate
Robot	UFACTORY xArm 6	Six-axis tabletop arm; official ROS 2 repository includes simulation, control, and MoveIt examples. Confirm repeatability and gripper control in Week 2.
Gripper	xArm parallel gripper	Simple grasp geometry; verify reliable disk-hole and outer-rim grasp before freezing board dimensions.
Camera	Intel RealSense D435i	The official xArm ROS 2 repository includes D435i model/link support. Confirm depth quality at the selected working distance.
Edge target	Jetson Orin NX 16GB	Practical Orin-class inference target. Use a supported carrier and 128GB+ NVMe. Profile before committing FoundationPose to the edge path.
Simulation host	Ubuntu x86_64, RTX 4080 16GB or better, 64GB RAM	Meets the current Isaac Sim 6 minimum GPU/VRAM class and provides headroom for a modest Isaac Lab task.
Safety	E-stop, fixed base, guarded workspace, speed limits	Mandatory before unattended or repeated physical trials.

Pinned software baseline

Layer	Baseline	Rationale
Simulation	Isaac Sim 6.x + compatible Isaac Lab	Use the latest supported patch within the pinned 6.x line after running the compatibility checker.
Robot + ROS	Ubuntu 22.04, ROS 2 Humble, xarm_ros2 humble branch	The official xArm repository lists this environment as tested.
Jetson	JetPack 6.2, Isaac ROS 3.2, ROS 2 Humble	This documented release line supports Jetson Orin and provides a coherent Orin-era baseline.
Planning	MoveIt 2 baseline; cuMotion variant	The core project remains completable if cuMotion integration is delayed or unsupported for the chosen model.
Packaging	Docker + lockfiles + recorded driver versions	Prevents an uncontrolled mix of incompatible 'latest' components.

Compatibility gate

No later than the end of Week 2: load the robot model, command the simulated gripper, acquire real RGB-D frames, execute one low-speed physical pick-and-place, and run one Isaac ROS package on the target compute. Failure triggers a documented hardware or version substitution before research development continues.

5. Experimental design

Comparisons, controlled disturbances, preregistered protocol, and explicit success criteria.

Systems under comparison

ID	System	Question answered
S0	Open-loop classical	What happens when commanded success is assumed?
S1	Visual verification only	How much value comes from detecting divergence without autonomous correction?
S2	Verification + rule recovery	How far can explicit reobserve/retry/replan rules go?
S3	Verification + residual policy	Does the learned bounded correction outperform S2 under alignment error?

Evaluation conditions

Condition	Disturbance	Purpose
E0	Nominal randomized starts	Establish unperturbed task reliability.
E1	Board-frame offset introduced within a preregistered millimeter range	Measure calibration robustness.
E2	Disk displaced after planning but before grasp or placement	Test divergence detection and replanning.
E3	Partial visual occlusion or low-confidence depth	Test abstention and reobservation behavior.
E4	Controlled grasp/placement misalignment	Isolate the residual policy's intended contribution.

Protocol

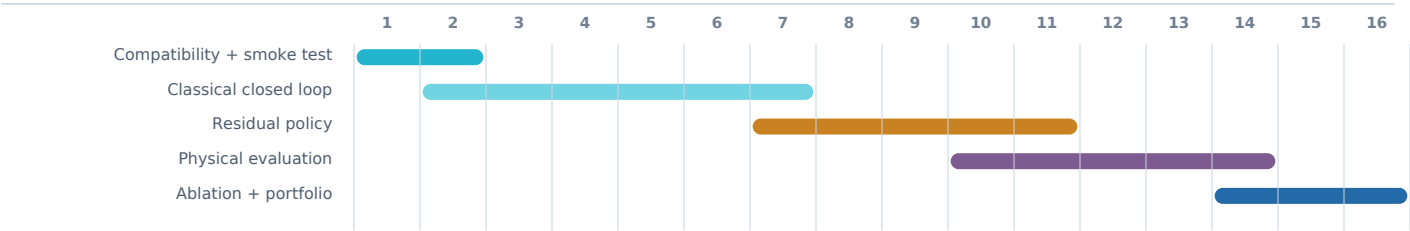
- Simulation: at least 1,000 episodes per system-condition cell across at least five held-out seeds.
- Physical: pilot first; then at least 30 full-puzzle trials per principal system condition, increasing the sample if power analysis indicates the expected effect cannot be resolved.
- Randomize system order within test blocks to limit warm-up, wear, lighting, and operator-order effects.
- Freeze disturbance ranges, thresholds, and exclusion criteria before the final battery.
- Report Wilson or bootstrap 95% confidence intervals, not percentages alone.
- Retain every failed episode; classify root cause without excluding inconvenient runs.

Primary endpoints and gates

Endpoint	Target / interpretation
Full-puzzle completion	Primary effect: S3 improves at least 25 percentage points over S0 under the preregistered disturbance mixture.
Illegal move / collision	Zero in the final battery; any event triggers investigation and rerun only after a documented fix.
Detection precision + recall	Target at least 90% each for predefined failure classes; report per-class confusion matrix.
Verification latency	Provisional p95 budget of 500 ms on target compute; freeze after Week 2 baseline.
Residual-policy gate	Must beat rule recovery on its intended E4 alignment condition without increasing safety stops or false corrections.
Sim-to-real gap	Report absolute and relative success degradation for each system; do not collapse to one aggregate number.

6. Sixteen-week execution plan

Milestones, explicit exit evidence, and go/no-go gates that protect the critical path.



Milestones

Weeks	Workstream	Exit evidence
1-2	Compatibility + hardware smoke test	Pinned environment; simulated arm/gripper; RGB-D capture; one low-speed physical pick-and-place; baseline latency report.
2-4	Symbolic planner + gross motion	A* produces only legal moves; 95/100 perfect-state simulated puzzle completions.
3-6	Perception + calibration	State reconstruction benchmark; board-frame drift measurement; documented symmetry treatment.
5-7	Verification + rule recovery	Expected/observed reconciliation; recovery from at least three fixed-seed disturbance classes.
7-11	Residual alignment policy	Training curves, held-out simulation results, abstention behavior, and S2-vs-S3 gate.
10-14	Physical evaluation	Pilot, frozen protocol, randomized final trials, structured logs, safety review.
14-16	Ablation + portfolio	Statistical report, profiling, failure analysis, public repository, demo video, and resume materials.

Go/no-go gates

Gate	Decision rule
G1 - Hardware	If the arm cannot repeatably place within the mechanical tolerance by Week 2, enlarge tolerance, change grasp strategy, or change arm before model development.
G2 - Perception	If geometry meets state accuracy and latency goals, keep it as primary; FoundationPose remains a benchmark rather than mandatory complexity.
G3 - Learning	If S3 does not beat S2 in held-out simulation by Week 11, ship the rule-based system and publish the negative learning result.
G4 - Edge	If the Jetson path misses memory or latency budget, retain x86 deployment and report the edge bottleneck instead of silently changing metrics.
G5 - Safety	Any collision, limit violation, or unexplained controller behavior pauses physical testing until root cause and mitigation are documented.

Minimum viable completion. A publishable portfolio result does not depend on the learned policy succeeding. The minimum viable project is S0-S2 with strong perception benchmarking, closed-loop recovery, physical trials, and honest GPU/CPU profiling.

7. Risks, observability, and reproducibility

A failure-aware research plan with run-level evidence and credible fallback paths.

Risk register

Risk	Early signal	Fallback
Arm repeatability	Placement scatter exceeds board tolerance	Increase hole/peg clearance, use visual servoing, or select a more repeatable arm.
Depth quality	Missing or noisy measurements near disk edges	Change camera angle/range, add controlled lighting, or use color/geometry fusion.
FoundationPose cost	Memory pressure or unacceptable p95 latency	Use geometric perception and reserve FoundationPose for offline comparison.
Sim-to-real failure	S3 improves simulation but not physical E4	Increase calibration/randomization fidelity; publish gap; retain S2.
ROS/version conflict	Driver or message incompatibility in Week 2	Pin documented branch/container or separate simulation and robot environments behind stable ROS interfaces.
Hardware delay	Delivery or integration misses Week 2	Continue fixed-seed simulation while sourcing a lab arm or compatible substitute.
Unsafe behavior	Limit, collision, or unexplained command	E-stop, safe home, reduced speed, root-cause review, and explicit reauthorization of the test.

Run-level observability

- Raw and overlay RGB-D frames, object/state estimates, confidence, and calibration snapshot.
- Planned symbolic move, nominal target transform, generated trajectory, and controller result.
- Expected-versus-observed discrepancy, failure classification, chosen recovery, and abstention reason.
- p50/p95 component and end-to-end latency, CPU/GPU utilization, memory, power mode, and thermal state.
- Immutable run ID, random seed, software/container digest, model hash, hardware configuration, and operator annotations.

Failure taxonomy

Layer	Examples
Perception	Missed disk, wrong identity/radius, wrong peg, bad height order, low confidence, frame drift.
Planning	No path, stale state, illegal action request, collision scene mismatch.
Execution	Missed grasp, slip, off-center placement, controller timeout, joint/velocity limit.
Recovery	False trigger, wrong correction, repeated retry, unsafe residual, failure to abstain.
Infrastructure	Frame drop, node crash, GPU OOM, thermal throttling, timestamp mismatch.

Repository and deliverables

Public repository with containerized setup; architecture decision records; simulation assets; ROS 2 packages; evaluation harness; frozen test manifest; raw and summarized results; safety protocol; technical report; three-to-five-minute demo; and a one-page NVIDIA-facing project brief.

8. Final demonstration and portfolio framing

A concise demonstration sequence that ends with evidence, not spectacle.

Demonstration sequence

- 1
- Show a randomized legal starting state and the live RGB-D/state overlay.
- 2
- Display the A* move plan and execute the first actions closed-loop.
- 3
- Introduce a preregistered disk displacement or placement misalignment.
- 4
- Show the expected/observed disagreement, confidence, and recovery decision.
- 5
- Execute the bounded correction or replan, then complete the puzzle.
- 6
- End with the actual benchmark: success intervals, failure matrix, p95 latency, and sim-to-real gap.

What the project should communicate

Signal	Evidence
Robotics systems engineering	A complete perception-planning-control-recovery loop with clean ROS 2 boundaries.
GPU judgment	Acceleration applied where measured value exists, with credible CPU/classical baselines.
Research maturity	Hypothesis, ablation, held-out evaluation, uncertainty intervals, and negative-result path.
Deployment competence	Pinned environments, Jetson profiling, driver/version documentation, and reproducible containers.
Safety and debugging	Bounded learning authority, failure taxonomy, structured telemetry, and go/no-go gates.

Resume bullet - use only after the results exist

Built and evaluated a closed-loop sim-to-real manipulation system using Isaac Sim, Isaac Lab, ROS 2, and NVIDIA-accelerated perception; added bounded visual recovery for peg alignment, deployed inference on Jetson, and improved seven-step physical task completion by [measured result] versus an open-loop baseline.

First ten working days

Day	Deliverable
1-2	Freeze candidate hardware, create compatibility matrix, run Isaac Sim checker, and pin containers.
3-4	Load xArm model, command joints/gripper in simulation, and implement legal symbolic state representation.
5-6	Acquire camera frames, estimate board frame, and measure depth quality at working distance.
7-8	Perform low-speed physical reach, grasp, and placement with safety observer and E-stop.
9-10	Publish compatibility report, baseline latency, first risks, and the final G1 decision.

The quality bar. The strongest outcome is not the robot completing Hanoi once. It is a defensible account of how accelerated perception, bounded learning, and closed-loop recovery affect real reliability - including the cases where they do not help.

9. References and decision record

Technical sources, compatibility notes, and the decisions to freeze after Week 2.

Primary technical references

[1] NVIDIA, Isaac Sim 6.0 system requirements. Minimum x86_64 GPU class listed as GeForce RTX 4080 with 16GB VRAM; accessed August 2026.

[2] NVIDIA, Isaac Lab. GPU-accelerated framework for robot learning and simulation; accessed August 2026.

[3] NVIDIA, Isaac ROS 3.2 Getting Started. Documents Jetson Orin, JetPack 6.1/6.2, and ROS 2 Humble compatibility; accessed August 2026.

[4] NVIDIA, FoundationPose. CAD-conditioned 6-DoF estimation and tracking; accessed August 2026.

[5] NVIDIA, Isaac ROS cuMotion. CUDA-accelerated motion generation and MoveIt 2 integration; accessed August 2026.

[6] UFACTORY, xarm_ros2. Official ROS 2 simulation, control, MoveIt, and D435i integration repository; accessed August 2026.

[7] NVIDIA, Jetson modules. Jetson Orin platform specifications and ecosystem; accessed August 2026.

Important compatibility note

NVIDIA's robotics software support matrix changes across releases. The reference stack intentionally distinguishes the current Isaac Sim host from an Orin-compatible Isaac ROS release line. At project start, record the chosen branch, test it on the actual hardware, and preserve the working environment. Do not upgrade during final evaluation without restarting qualification.

Decision record to complete after Week 2

Decision	Final selection	Evidence
Robot + gripper	_____	Repeatability, control, safety, and ROS 2 smoke test
Camera + mount	_____	Depth quality, calibration stability, and occlusion test
Simulation host	_____	Compatibility checker, scene FPS, and training throughput
Edge target	_____	Memory, p50/p95 inference, power, and thermals
Perception primary	_____	Geometry vs FoundationPose benchmark
Motion primary	_____	MoveIt 2 vs cuMotion feasibility and planning benchmark
Residual-policy gate	_____	S2 vs S3 held-out simulation result

Hanoi-RX research project / Research scope frozen around visual verification, bounded residual alignment, and empirical sim-to-real reliability.